# Contents



# Papers that share datasets — train vs not

Campus account is enough to open PDFs and download files. Automation only **finds titles**. Do not scrape IEEE / ACM / ScienceDirect.

Re-run: `python scratch/find_shared_datasets.py`

Links HTTP-checked 2026-08-15 (real records; ACM/IEEE may be gated). Tip: Zenodo record titles often match the paper name — search that title on arXiv / ACM / IEEE to fill the Paper column.

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## Train the model (IsolationForest / KMeans)

The earlier "only two" were **corpus names**. RCAEval alone is already **three systems** (Online Boutique, Sock Shop, Train Ticket) and 735 labeled inject cases.

Native ChaosGen features still come from **your lab**. Everything else needs a **column map** (metrics/logs/traces → windows) and labeled healthy vs fault. Never mix load-only traffic into the failure class.

| Item | Architecture | Paper | Dataset | How to train |
|---|---|---|---|---|
| **Your lab — healthy vs serious inject** | whatever you deploy (boutique / k8s / …) | — | (your stack) | **Required.** Baseline vs network-cut / node-kill / connection wipe. |
| RCAEval (RE1–RE3) | **Microservices ×3:** Boutique, Sock Shop, Train Ticket | ACM 10.1145/3701716.3715290 | Figshare · Zenodo 14590730 · GitHub | Best published match. CPU/mem/disk/delay/loss/socket + code faults; `inject_time.txt` labels. |
| Nezha `rca_data` | Microservices: Boutique + Train Ticket | ACM 10.1145/3611643.3616249 (FSE 2023) · ResearchGate | GitHub · Zenodo 8276375 | Fault-free vs fault-suffering metrics/logs/traces + fault_list labels. |
| Eadro | Microservices: Train Ticket + **DeathStarBench SocialNetwork** | arXiv 2302.05092 (ICSE 2023) · ResearchGate | Zenodo 7615393 · 7615394 | Injected CPU exhaustion, delay, packet loss; traces+logs+KPIs. |
| TrainTicketTra | Microservices: Train Ticket | IEEE Xplore 11500150 | Zenodo 17811971 | Clean vs **nine seeded-fault versions**; traces+metrics+logs. |

| Item | Architecture | Paper | Dataset | How to train |
|---|---|---|---|---|
| LO2v2 | Microservices: **LightOAuth2** (OAuth / identity, not a demo shop) | arXiv 2504.12067 | Zenodo 18937117 | Prometheus/cAdvisor + logs; tests tagged **correct vs error**. Start with `LO2v2-metrics.zip` (~1.2 GB), not the 62 GB raw. |
| Open5GS | **Cloud-native 5G core** on Kubernetes | TU Delft thesis PDF | 4TU | Prom+Loki+Jaeger; baseline/fault/recovery. Telco, not boutique. |
| Cloud-OpsBench | **Kubernetes** control-plane + Boutique / Train Ticket | arXiv 2603.00468 | GitHub | 754 cases, `metrics.csv` + logs + k8s snapshots, 57 fault types (admission/scheduling/runtime/…). Map metrics; ignore agent chat traces for IF. |
| PetShop | Distributed **multi-service** shop | arXiv 2311.04806 | GitHub | 68 injected perf issues; latency/availability. Adapter required (not Prom/Loki). |
| Loghub HDFS_v1 + HDFS_v3 | **Distributed FS / client–server** (HDFS) | arXiv 2008.06448 (ISSRE 2023) | GitHub · Zenodo 8196385 | v1: labeled normal/anomaly traces. v3 TraceBench: **17 injected faults** vs normal. Log/trace features, not PromQL. |
| **GAIA** | MicroSS simulation: **10 microservices** + MySQL/Redis | IEEE TSC 2023 | GitHub | Metrics + logs + traces; five injected faults (stuck, crash, login, file missing, access denied) with injection records. Map to windows; do not mix with lab PromQL blindly. |
| **AIOps Challenge 2020** | Real production **MSS** (Zhejiang Mobile) | IEEE 9293310 · survey context ACM 10.1145/3715005 | GitHub (community dump) | Labeled anomaly / RCA cases (metrics + call chains). Best published *production* set. Repo is often an index; pull the actual dump from the README links. Schema boutique Prom. |
| **MultiDimension-Localization** | Production **21 microservices** | Sensors 2025 | GitHub | Multivariate metrics only (19 per service); 58 labeled cases (contention, latency, app errors, dependency). Easier adapter than full traces. |

**Held — do not train yet:** TrainTicket-DéjàVu — paper ACM 10.1145/3540250.3549092 / arXiv 2207.09021 · dataset Zenodo 6955909 — confirm healthy-vs-fault columns before training. SockShop knowledge-graph benchmark — paper IEEE TNSM / IEEE Xplore / ResearchGate; open-source framework in the paper (no stable dump URL in the survey row); RCAEval already covers Sock Shop inject.

**Search is enough to stop.** Zhang et al.'s TOSEM survey (ACM 10.1145/3715005) already catalogs most of the same public dumps (GAIA, AIOps Challenge, TrainTicket variants, Nezha, Eadro, …). Overlap with this note is expected — it is confirmation, not a miss. Post-survey additions here (RCAEval, LO2, Open5GS, Cloud-OpsBench, PetShop, MultiDimension) still leave the same **architectural** holes: no public serverless/FaaS *injected-fault* dump, no clean monolith Prom inject dump. More paper search will not invent those. Ask the advisor whether this corpus + **lab inject on the thesis stack** is sufficient; do not keep hunting unless they name a missing architecture.

Local clones (gitignored, do not email): `data/public-datasets/`. Submit **this note**, not the raw dumps.

**Still scarce (do not invent labels):** public **serverless FaaS** dumps with *injected faults* (OpenWhisk/Lambda) are basically missing. Azure Functions 2019 / KSWD are **workload shapes**, not failures — do not train as the anomaly class. Cover "cloud-native / k8s" via Cloud-OpsBench + Open5GS instead.

**Monolith:** no clean public Prom inject corpus showed up. Closest: HDFS (distributed FS) and 5G core (many tightly coupled NFs). Your own lab can still run a modular monolith and inject.

Do **not** treat any published set as a drop-in replacement for lab inject.

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## Thesis support — do not train

Cite, compare, or offline RCA eval. Wrong object or wrong schema for ChaosGen fit.

| Item | Paper | Dataset | Why not train |
|---|---|---|---|
| RCAEval paper | ACM 10.1145/3701716.3715290 | (see TRAIN row) | PDF citation only |
| Online Boutique telemetry | Zenodo record (dataset deposit; no separate journal paper found) | Zenodo 20685204 | Load replay, **not** disaster |
| OpenTelemetry AIOps | Zenodo companion (IEEE Access submission titled in the deposit) | Zenodo 19462084 | Anomaly benchmark, not chaos inject |
| TORAI RCA | arXiv 2604.13522 | Figshare 31925976 | Call-graph RCA, not collector windows |
| LEMMA-RCA (IT slices only) | arXiv 2406.05375 | lemma-rca.github.io | RCA labels; ignore SWaT/WADI water OT |

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## Not useful — do not train

| Item | Paper | Dataset | What to do |
|---|---|---|---|
| CATS | Zenodo deposit (CATS; cites PVLDB AD survey) | Zenodo 8338435 | Ignore — generic synthetic series |
| GWDG GPU telemetry | arXiv 2603.28781 | Zenodo 19052366 | Ignore — wrong layer |
| VECROsim | IEEE ISSRE 2022 | Figshare 30627446 | Ignore unless README proves otherwise |
| ARISE | Zenodo deposit (paper under review per deposit) | Zenodo 21096887 | Ignore for ML — incident RAG |

| Item | Paper | Dataset | What to do |
|---|---|---|---|
| Cloud Uptime Archive | IEEE TPDS | GitHub | Cite only |
| Failure Trace Archive | CCGrid 2010 | — | Cite only |
| CFDR | USENIX CFDR | usenix.org/cfdr-data | Cite only |
| Alibaba cluster | arXiv 1808.02919 | GitHub | Cite only |
| Google Borg traces | Borg EuroSys 2015 | GitHub | Cite only |
| AIOps KPI benchmarks | arXiv 2208.03938 | — | Cite only |
| Distributed-trace SMS package | Zenodo replication package (SMS title in deposit; no separate DOI found yet) | Zenodo 21772871 | Reading list only |